

Chapter 5 - Room Survey and Data Collection

ROOM SURVEY AND DATA COLLECTION

5.1 Room Survey

A physical survey of the HOD cabin was carried out to collect the actual data required for cooling load calculation. The survey included room dimensions, building construction, doors, windows, openings, occupants, lighting and electrical equipment.

The collected observations provide the basic input for the cooling load calculations presented in Chapter 6.

5.2 Room Dimensions

The dimensions of the HOD cabin were measured using a measuring tape.

Parameter	Value
Length	_____
Breadth	_____
Height	_____
Floor Area	_____
Room Volume	_____

The floor area and volume are calculated using:

$$\text{Floor Area} = \text{Length} \times \text{Breadth}$$

$$\text{Room Volume} = \text{Length} \times \text{Breadth} \times \text{Height}$$

These values are used for determining heat transfer and air-related cooling loads.

5.3 Building and Room Conditions

The HOD cabin is located on the ground floor of the college building. Classrooms are present above the cabin, while the floor is directly above the ground level.

The following conditions were observed:

Walls enclose the cabin on the required sides.

The main entrance door is generally kept open.

Windows are provided for light and air movement.

Small netted openings remain open.

A separate door connects the cabin to the private toilet.

Tables, chairs and cupboards are present inside the cabin.

These conditions are considered while identifying the different sources of cooling load.

5.4 Doors, Windows and Openings

The doors, windows and other openings were inspected and their dimensions were recorded.

Opening	Quantity	Dimensions
Main door	_____	_____
Toilet door	_____	_____
Windows	_____	_____
Small netted openings	_____	_____

The main door and small open/netted openings allow outdoor air to enter the cabin. Therefore, they are considered while determining ventilation and infiltration load.

5.5 Occupants and Internal Heat Sources

The number of occupants normally present in the cabin was observed during the survey. Occupants contribute both sensible and latent heat to the room.

Other internal heat sources identified during the survey include:

Lights

Fans

Computer and monitor

Other electrical equipment

The heat generated by these sources contributes mainly to the sensible cooling load.

5.6 Lighting and Electrical Equipment

The quantity and power rating of the major electrical equipment were recorded.

Equipment	Quantity	Power
Lights	_____	_____ W
Fans	_____	_____ W
Computer	_____	_____ W
Monitor	_____	_____ W
Other equipment	_____	_____ W

For electrical equipment, the heat gain can be estimated from its power consumption:

$$\text{Heat Gain} = \text{Number of Equipment} \times \text{Power per Equipment}$$

The operating condition of the equipment is also considered during the cooling load calculation.

The measured dimensions and observed conditions obtained from the room survey provide the necessary data for calculating conduction, solar, internal, ventilation and infiltration heat gains. These values are used in Chapter 6: Actual Cooling Load Calculation.